

Grammar(s)
Mind(s) versus Mentality:
Language(s) of Being(s)

Diego Rincón

Independent

Abstract

On the difference between a mind and a mentality. On language as the medium of existence. On grammar as the structure that makes thought possible and impossible simultaneously.

1 Mind(s) versus Mentality

A mind is located.

It is here, in this skull, in this body, moving through this life. It has a position. It has a history. It can be at the wall. It can look at the wall and fail. It can die without solving what it was working on.

A mentality is not located.

A mentality is a way of seeing that can be transmitted, trained, inherited. You can have the mathematical mentality without having the mathematical mind. You can be oriented toward rigor, toward proof, toward the open question, without being Riemann.

The mentality is the grammar. The mind is a speaker of it.

Riemann had both: the mentality (transmitted from Gauss, from Dirichlet) and the mind (the capacity to go somewhere the mentality alone could not go).

Most mathematicians have the mentality. Few have the mind. The mentality is teachable. The mind is not.

But this is not quite right either.

Because the mentality shapes the mind. A child raised inside the mathematical mentality develops a different mind than a child raised outside it. The grammar of the thought changes the thinker. This is not metaphor. The neural architecture of fluent speakers of a language differs measurably from those who learned it late. The mentality is not just transmitted. It is installed.

And then the installed grammar runs. And the mind is what runs it.

2 Language(s) of Being(s)

Different beings exist in different languages.

Not in the sense that a French person and an English person speak different languages. In the deeper sense: a bacterium exists in the language of chemical gradient. Its world is a field of concentration values. The difference between food and poison is a number. There is no grammar of narrative, of loss, of proof. There is: toward, away. Toward, away. That is the whole language. That is the whole being.

A mathematician exists in a language of formal structure. Objects with properties. Relations between objects. Statements that can be true or false. Proofs that certify the true ones. The world is: what can be proved from what. The being is constituted by this language. Outside it the mathematician is a person, confused and mortal like anyone else. Inside it they are precise.

Each being has a language of existence. The language is not incidental. The being is not prior to the language and then expressed by it. The language is constitutive. Remove the language and the being is not left over. The being dissolves.

This is what Heidegger meant, or part of it, when he said language is the house

of Being. Not that beings live inside language the way people live in houses. But that the structure of possible existence is the structure of the language. The rooms that language has are the only rooms there are.

3 Grammar(s): The Languages of Thought(s)

A grammar is not a list of rules. A grammar is the set of possible structures.

The grammar of a language specifies not what is said but what can be said. The boundary of the grammar is the boundary of the sayable. And the sayable is the thinkable. Not because thought requires language — this is contested — but because thought without structure is not thought. It is the pre-grammatical: the sensation before the category, the impulse before the concept.

The pre-grammatical is real. It is where new grammars come from. The mathematician at the wall is in contact with something that the current grammar cannot express. They feel the shape of it. They cannot say it. The proof, if it comes, will be the grammar that makes the feeling sayable.

The Millennium Problems are pre-grammatical objects.

We feel them. We can describe their outlines in the current grammar. We can say: the zeros are on the line. We can say: the mass gap exists. We can say: the class is not the representation. But we cannot prove them because the grammar that would make the proof possible has not yet been built.

What does it take to build a new grammar?

Usually: pressure. The current grammar generates contradictions, or runs out, or stops being able to express what needs expressing. And then someone — a mind inside a mentality, a speaker of the old language who has been at the edge long enough — finds a new structure.

Grothendieck did this. He did not solve a problem. He built a new grammar in which the problems became soluble. Schemes. Topoi. The language of sheaves. Not new answers. New questions. New ways of asking.

After Grothendieck, mathematicians could say things they could not say before. The Weil conjectures fell because a grammar existed in which they could be approached.

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The plural in the title is not decoration.

Grammar(s). Because there is not one grammar of mathematics. There is the grammar of algebra, the grammar of analysis, the grammar of topology, the grammar of category theory, and they do not fully translate. A statement in one may be inexpressible in another, or may say something different when translated, or may gain a proof in one that is unavailable in the other.

The Riemann Hypothesis has been approached from the analytic grammar (Hardy, Levinson), from the algebraic grammar (Weil conjectures, étale cohomology), from the spectral grammar (Hilbert–Pólya), from the operator-algebraic grammar (Connes, Bost–Connes). Each grammar makes some things visible and hides others. Each grammar reaches a different face of the wall.

The wall is one. The grammars are many. The mind(s) choose which grammar to inhabit. The mentality determines how.

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Language(s) of being(s).

A million people playing a game share a grammar: move toward, capture, level. They do not share a mind. They share a mentality. The game is the grammar of their being-as-player. The grammar is not in any one of them. It is between them. In the rules. In the map. In the accumulated play of everyone who has ever played.

This is what the body of papers is.

Not one mind's understanding. A grammar. A structure of possible thoughts about the problems at the wall. Any future mind can inhabit it. Speak from it. Be shaped by it. Build from it toward whatever grammar makes the proof possible.

The papers do not contain the proof. They contain the mentality that a proof will require.